```
import pandas as pd

iris_df = pd.read_csv('/content/Iris.csv')

print("First 5 rows of the Iris dataset:")
print(iris_df.head())

print("\nColumn names and their data types:")
iris_df.info()
```

```
First 5 rows of the Iris dataset:
   Id  SepalLengthCm  SepalWidthCm  PetalLengthCm  PetalWidthCm      Species
0   1            5.1           3.5            1.4           0.2  Iris-setosa
1   2            4.9           3.0            1.4           0.2  Iris-setosa
2   3            4.7           3.2            1.3           0.2  Iris-setosa
3   4            4.6           3.1            1.5           0.2  Iris-setosa
4   5            5.0           3.6            1.4           0.2  Iris-setosa

Column names and their data types:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 6 columns):
 #   Column         Non-Null Count  Dtype  
---  ------         --------------  -----  
 0   Id             150 non-null    int64  
 1   SepalLengthCm  150 non-null    float64
 2   SepalWidthCm   150 non-null    float64
 3   PetalLengthCm  150 non-null    float64
 4   PetalWidthCm   150 non-null    float64
 5   Species        150 non-null    object 
dtypes: float64(4), int64(1), object(1)
memory usage: 7.2+ KB
```

```
from sklearn.model_selection import train_test_split

X = iris_df.drop(['Id', 'Species'], axis=1)
y = iris_df['Species']


X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

print("Shape of X_train:", X_train.shape)
print("Shape of X_test:", X_test.shape)
print("Shape of y_train:", y_train.shape)
print("Shape of y_test:", y_test.shape)
```

```
Shape of X_train: (120, 4)
Shape of X_test: (30, 4)
Shape of y_train: (120,)
Shape of y_test: (30,)
```

```
from sklearn.linear_model import LogisticRegression


model = LogisticRegression(max_iter=200, random_state=42)


model.fit(X_train, y_train)

print("Logistic Regression model trained successfully.")
```

```
Logistic Regression model trained successfully.
```

```
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix


y_pred = model.predict(X_test)


accuracy = accuracy_score(y_test, y_pred)
print(f"Model Accuracy: {accuracy:.4f}")


print("\nClassification Report:")
print(classification_report(y_test, y_pred))
```

```
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))
```

```
Model Accuracy: 1.0000

Classification Report:
                 precision    recall  f1-score   support

    Iris-setosa       1.00      1.00      1.00        10
Iris-versicolor       1.00      1.00      1.00         9
 Iris-virginica       1.00      1.00      1.00        11

       accuracy                           1.00        30
      macro avg       1.00      1.00      1.00        30
   weighted avg       1.00      1.00      1.00        30


Confusion Matrix:
[[10  0  0]
 [ 0  9  0]
 [ 0  0 11]]
```

```
import matplotlib.pyplot as plt
import seaborn as sns

cm = confusion_matrix(y_test, y_pred)

plt.figure(figsize=(8, 6))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues',
            xticklabels=model.classes_, yticklabels=model.classes_)
plt.title('Confusion Matrix')
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.show()
```